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Membrane depolarization drives tumor-suppressive cell competition via Hedgehog signaling activation

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Abstract

Tumor-suppressive cell competition (TSCC) is an evolutionarily conserved process that safeguards tissue integrity by selectively eliminating less-fit, precancerous cells. While bioelectric regulation via plasma membrane potential (V_m) is emerging as a key modulator of cellular fitness, its mechanistic role in TSCC remains underexplored. Here, we combine *Drosophila* genetics, single-cell RNA sequencing (scRNA-seq), and mammalian co-culture models to indicate that V_m depolarization acts as a critical driver of TSCC through Hedgehog (Hh) signaling. In *Drosophila* eye epithelia, *scribble*-deficient (*scrib*^{-/-}) “loser” clones exhibit mitochondrial respiratory chain defects, leading to reduced ATP synthesis and subsequent plasma membrane depolarization. This depolarization stabilizes Smoothened at the membrane, aberrantly activating Hedgehog signaling and triggering the elimination of *scrib*^{-/-} clones. Notably, we observe a striking mechanistic parallel in mammalian epithelia: *Scrib*-depleted Madin-Darby Canine Kidney (MDCK) cells similarly undergo V_m depolarization-dependent elimination associated with Hh pathway activation. Together, our work advances the understanding of bioelectricity in cancer surveillance and suggests that targeting membrane potential could offer a promising therapeutic strategy for cancer prevention by exploiting cell competition mechanisms.

Keywords

Drosophila; Membrane potential; Scribble; Cell competition; Hedgehog signaling; Tumor suppression; Single-cell RNA sequencing

Introduction

Cell competition is an evolutionarily conserved fitness-sensing mechanism by which cellular fitness is assessed, leading to the selective elimination of less fit “loser” cells while favoring the survival and proliferation of fitter “winner” populations (Madan et al., 2022; van Neerven and Vermeulen, 2023). Originally described in *Drosophila melanogaster* wing discs, this process was first observed when cells heterozygous for ribosomal mutations were outcompeted by their wild-type (WT) neighbors (Morata and Ripoll, 1975).

Tumor-suppressive cell competition (TSCC), a specialized form of cell competition, acts as a frontline defense against tumor initiation by selectively removing precancerous cells harboring oncogenic mutations or fitness defects (Kanda and Igaki, 2020; Ayukawa et al., 2021). During TSCC, compromised “loser” cells, such as those lacking tumor suppressor genes like *scribble* (*scrib*), are actively outcompeted by neighboring healthy “winner” cells through numerous mechanisms (Norman et al., 2012; Feigin et al., 2014; Kanda and Igaki, 2020; Dillard et al., 2021; Kong et al., 2022). Similarly, mammalian epithelial cells employ a comparable strategy, known as epithelial defense against cancer (EDAC), to detect and eliminate neighboring cells that exhibit early signs of oncogenic transformation (Norman et al., 2012; Yako et al., 2018; Tanimura and Fujita, 2020). These evolutionarily conserved processes are vital for maintaining tissue integrity, yet the molecular pathways and biophysical drivers that translate cellular fitness disparities into elimination signals remain incompletely understood.

Plasma membrane potential (V_m), the electrical gradient across cellular membranes, plays a fundamental role in regulating electrical activity in excitable cells such as neurons (Glynn, 2002). V_m is established by the unequal distribution of ions across the membrane, facilitated by ion channels and transporters with specific selectivity and permeability. Notably, accumulating studies have highlighted the significance of V_m shifts, particularly membrane depolarization (a loss of charge differential across the membrane), in non-excitable cells, including cancer cells. Depolarization has been linked to altered expressions of ion channels and pumps, as well as oncogenic transformation (Goldman, 1943; Hodgkin and Huxley, 1945; Farias et al., 2004; Levin, 2007; Prevarskaya et al., 2010; Chernet and Levin, 2013; Yang and Brackenbury, 2013). For instance, in oncogene-expressing *Xenopus* cells and tumorigenic mammalian cell lines, depolarized V_m has been shown to promote cancer-like phenotypes in these systems (Chernet and Levin, 2013; Levin, 2021). However, validating the translational

relevance of these findings requires further investigation in primary human tissues, *in vivo* mammalian models, and lineage-traced precancerous epithelia.

Mitochondrial dysfunction further exacerbates this bioelectric imbalance: impaired electron transport chain (ETC) activity reduces ATP synthesis, compromising Na⁺/K⁺-ATPase function and further depolarizing the membrane (Lee et al., 2011; Spannl et al., 2020). As mitochondria regulate both metabolic flux and apoptotic signaling, their dysfunction critically influences competitive cellular fitness (Green and Kroemer, 2004; Lima et al., 2021). Although these findings underscore the importance of bioelectricity in cancer biology, the role of V_m dynamics in the context of TSCC remains unexplored.

In this study, leveraging single-cell RNA sequencing (scRNA-seq), *Drosophila* models of *scrib*-deficiency-induced TSCC, and mammalian co-culture systems, we uncover a mechanism by which mitochondrial respiratory defects in *scrib*-depleted cells lead to reduced ATP levels, membrane depolarization, and stabilization of Smoothened, thereby aberrantly activating Hedgehog (Hh) signaling to drive TSCC. Notably, the observation of similar bioelectric and signaling alterations in our mammalian co-culture systems suggests that this regulatory axis operates through a parallel mechanism. Our findings establish bioelectric regulation as a crucial determinant of competitive cellular fitness and provide a mechanistic foundation for harnessing membrane potential modulation in cancer surveillance and therapeutic strategies.

Results

Single-cell RNA-seq profiling of wild-type and *scrib* mutant clone-bearing eye-antennal discs

Traditional bulk transcriptomic approaches have provided limited resolutions into the dynamics of cell competition, as they mask the heterogeneity of cellular responses and fail to capture rare or transient cell states. scRNA-seq emerges as a transformative tool to dissect this process at unprecedented resolution (Kolodziejczyk et al., 2015). To comprehensively understand the transcriptional changes specific to *scrib*-deficient cells during TSCC, we performed scRNA-seq on *Drosophila* eye-antennal imaginal discs bearing GFP-labeled WT or *scrib*-mutant clones using the Singleron Matrix® Single Cell Processing platform (Fig. 1A).

Following stringent quality control, filtering, and doublet removal steps, we retained 90,217 high-quality cells, with a median of 2,221 unique molecular identifiers (UMIs) and 695 expressed genes per cell (Fig. S1A). Uniform Manifold Approximation and Projection (UMAP) analysis revealed distinct cell clusters corresponding to all expected cell identities within the larval eye disc (Figs. 1B and S1B).

Using the expression patterns of known marker genes, we assigned putative spatial identities to the scRNA-seq clusters as an inferential annotation. In the antennal compartment, outer antennal rings (A1 and A2 clusters) were characterized by the expression of *hth* and *ct* (Bravo González-Blas et al., 2020) (Fig. 1C), while inner rings (A3 and arista) expressed *toe* (Fig. 1C), a gene restricted to medial/distal antennal segments and the dorsal-ventral boundary anterior to the morphogenetic furrow (MF) (Yao et al., 2008). Within the eye disc, the MF cluster was identified via *E(spl)* genes and *rn* expression (de Celis et al., 1996; Raja et al., 2024) (Fig. 1C). The peripodial membrane domain, anterior to the MF, expressed *CG17278* and *bnb* (Fig. 1C), consistent with prior reports (Ariss et al., 2018). Photoreceptor precursors and progenitors (Pro&Pre) were marked by *toy* (Fig. 1C), a gene specific to undifferentiated cells anterior to the MF (Czerny et al., 1999). The head vertex (HV) cluster expressed *Ance* and *sog* (Ariss et al., 2018). Posterior photoreceptor clusters were distinguished by *Mmp2* and *twit*, while late photoreceptors (LPR) expressed *retn* and *ru* (Fig. 1C). Interommatidial cells were identified by *rgn*, *sls*, and *Gasp* expression (Ariss et al., 2018) (Fig. 1C). The second mitotic wave (SMW) cluster, representing a subset of cells in the posterior region undergoing an additional round of division, exhibited the expression of cell cycle genes such as *Mcm2* and *PCNA* (Ariss et al., 2018; Bollepogu Raja et al., 2023) (Fig. 1C). Notably, a population anterior to the MF displayed elevated *Ldh* and *Ald1* expression, aligning with reported glycolytic flux increases linked to *Rbf* mutations in this region (Ariss et al., 2018). Two glial subtypes, perineurial glia (PG) and wrapping/subperineurial glia (WG&SPG), were resolved by *Act57B*, *trol*, and *Gabat* expression (Ariss et al., 2018), while hemocytes were defined by *CG4250* and *He* expression (Ariss et al., 2018) (Fig. 1C).

Collectively, these results demonstrate that our scRNA-seq dataset robustly recapitulates the known cell-type diversity in the larval eye-antennal disc, providing a solid foundation for investigating transcriptional changes associated with *scrib* loss during TSCC.

***scrib* mutant loser clones exhibit mitochondrial ETC deficiency**

Next, to further investigate the unique transcriptional landscape of *scrib* mutant clones, we removed glia cells and hemocytes, leveraging GFP as a marker to isolate *scrib*^{-/-} cells from a pool of 41,565 cells (Fig. 1D). Clusters 16 and 21 were identified as distinct *scrib*^{-/-} populations based on GFP expression (Figs. 1D, S1C, S1D). We subsequently inferred the spatial origins of these clusters by mapping their transcriptional profiles against known marker genes for eye disc development. Cluster 16 corresponds to cell types located anterior to the MF, including head vertex, EAB&A1, and EPR cells. In contrast, cluster 21 consists predominantly of interommatidial cells (INT), which are situated posterior to the MF (Fig. S1E).

Consistent with the hyperactivation of JNK signaling resulting from *scrib* loss, both clusters exhibited co-expression of GFP and increased expression of the Jun N-terminal kinase (JNK) pathway targets, including *Mmp1* and *scaf* (Uhlirova and Bohmann, 2006; Rousset et al., 2010) (Fig. 1D). This JNK activation served as a specific positive control, validating the identity of the isolated *scrib* mutant cells. While we also observed enrichment of other canonical pathways (e.g., Hippo and JAK-STAT) previously reported to be dysregulated in *scrib* mutants (Table S1), a detailed analysis of these pathways is beyond the scope of this study.

Next, differential gene expression analysis between *scrib*^{-/-} clusters and WT cells revealed that upregulated genes were enriched in Gene Ontology (GO) terms related to mitochondrial oxidative phosphorylation (OXPHOS) and the respiratory ETC (Fig. 1E). To evaluate OXPHOS complex integrity, we quantified the average expression of different subunits of mitochondrial complexes I-V (Vercellino and Sazanov, 2022). Remarkably, *scrib*^{-/-} clusters demonstrated a pronounced reduction in the expression of core subunits of complexes I (NADH dehydrogenase) and IV (cytochrome c oxidase) (Figs. 1F, S1F-S1I). Conversely, non-core components were upregulated (Figs. 1F, S1F-S1I), likely reflecting compensatory mechanisms aimed at preserving overall ETC efficiency (Pulliam et al., 2014).

Given the high statistical enrichment of mitochondrial signatures and their direct mechanistic link to our subsequent functional validations (e.g., ATP levels), we prioritized mitochondrial ETC deficiency for further investigation. Although scRNA-seq revealed a prominent transcriptional signature of ETC deficiency in *scrib*^{-/-} cells, we sought to determine whether this deficiency directly drives the “loser” phenotype through spatially resolved functional assays in intact tissues. Consistent with reduced mitochondrial activity, *scrib*^{-/-} clones exhibited a significant decrease in mitochondrial membrane potential ($\Delta\Psi_m$), as assessed by

tetramethylrhodamine methyl ester (TMRM) staining (Figs. 1G and S1J). Furthermore, to examine potential changes in mitochondrial mass, we performed additional immunofluorescent staining for the ATP synthase F1 subunit alpha (ATP5 α) and detected no significant difference in ATP5 α levels between WT and *scrib*^{-/-} clones (Fig. S1K). Given the central role of mitochondria in energy production, we next measured *in vivo* ATP levels using ATP-SPARK, a reporter system adapted from the separation of phases-based activity reporter of kinase (SPARK) technology (Zhang et al., 2018). The ATP-SPARK probe consists of an ATP-binding domain fused to the SPARK reporter module. Upon binding intracellular ATP, the probe undergoes a conformational change that triggers liquid-liquid phase separation, resulting in the formation of distinct fluorescent condensates (droplets). This mechanism allows for the visual detection of ATP availability, where relative ATP levels are quantified by counting the number of fluorescent droplets per unit area within the clones. Compared to controls, *scrib*^{-/-} loser clones displayed a marked reduction in ATP production (Fig. 1H), whereas clonal overexpression of *yki* or oncogenic *Ras* (*Ras*^{V12}), both of which confer a benign tumor-like growth advantage, exhibited elevated ATP-SPARK activity (Fig. S1L). Furthermore, mitochondrial dysfunction disrupted calcium homeostasis, as evidenced by elevated cytosolic Ca²⁺ levels detected using the Fluo-4 AM calcium ion fluorescent probe (Fig. S1M).

To determine whether the mitochondria defects observed in *scrib*^{-/-} clones are strictly dependent on the mosaic context due to competitive interactions with WT neighbors or are driven by intrinsic, cell-autonomous stress, we examined mitochondrial phenotypes under non-mosaic conditions. We uniformly knocked down *scrib* across large epithelial compartments in both eye-antennal and wing discs using the *GMR-Gal4* driver (posterior to the MF) and the *nub-Gal4* (wing pouch), respectively. Under these homotypic conditions, widespread *scrib* depletion did not result in significant alterations in $\Delta\Psi$ m (Fig. S1N and S1O). These findings indicate that the mitochondrial defects observed in *scrib*^{-/-} clones are not solely caused by intrinsic cellular stress but arise specifically in mosaic tissues, likely as a consequence of competitive interactions with surrounding WT cells.

Collectively, these data indicate that mitochondrial ETC activity is impaired in *scrib* mutant loser cells during TSCC (Fig. 1I).

Membrane potential dynamics regulate tumor-suppressive cell competition

ATP serves as the primary energy currency in eukaryotic cells, fueling diverse processes such as ion transport via the Na^+/K^+ ATPase (sodium pump). As a major consumer of ATP, Na^+/K^+ ATPase relies on a steady ATP supply to maintain electrochemical gradients critical for cellular excitability, osmotic balance, and signal transduction (Glynn, 2002) (Fig. 2A). Notably, we observed a marked reduction of $\text{ATP}\alpha$, the catalytic subunit of the Na^+/K^+ ATPase, at both the protein and transcriptional levels in *scrib* mutant clones (Fig. 2B and 2C). To determine whether this reduction in $\text{ATP}\alpha$ represents a general feature of cell competition, we extended our investigation to additional cell competition models. Intriguingly, $\text{ATP}\alpha$ downregulation was specific to clones with mutations in cell polarity regulators, including *scrib* and *lethal (2) giant larvae (lgl)*, but absent in *myc*-deficient competitive contexts (Fig. S2A and S2B). This finding reveals a previously unrecognized link between polarity loss and $\text{ATP}\alpha$ suppression, a correlation distinct from *myc*-driven competition, though the underlying molecular mechanism and functional consequences remain to be fully elucidated.

Concurrently, we observed a significant upregulation of the γ subunit of the Epithelial Sodium Channel (ENaC) in *scrib* mutant clones (Fig. 2D), indicating a potential increase in sodium transport capacity (Kellenberger and Schild, 2002). Given that Na^+ enters epithelial cells via ENaC and is extruded by Na^+/K^+ -ATPase into the interstitial space, the combination of ENaC γ upregulation and impaired Na^+/K^+ -ATPase function may synergistically contribute to plasma membrane depolarization (Morth et al., 2011; Lingemann et al., 2019; Spann et al., 2020). To test this hypothesis, we measured V_m in *scrib* mutant clones using the anionic potentiometric probe DiBAC4(3) (Adams and Levin, 2012). Consistently, *scrib*^{-/-} clones exhibited increased DiBAC4(3) uptake (Fig. 2E), whereas global *scrib* depletion did not significantly alter V_m (Fig. S2C and S2D), indicating that membrane depolarization occurs specifically in “loser” cells engaged in competitive interactions with WT “winner” cells.

To establish a causal link between V_m dynamics and *scrib*-mediated TSCC, we genetically manipulated depolarization levels. Knockdown (KD) of the sodium channel *ripped pocket (rpk)* reduced depolarization and significantly suppressed the elimination of *scrib*^{-/-} clones (Figs. 2F and S2E). Notably, apoptosis induced by *scrib* loss, assessed by the activation of the effector caspase Death caspase-1 (Dcp-1), remained unchanged upon *rpk* KD (Fig. S2F and S2G), indicating that the rescue of clone elimination is independent of apoptotic modulation. To further evaluate whether V_m influences apoptosis outside the context of competition, we utilized a well-established model of cell death involving ectopic expression of the $\text{TNF-}\alpha$

homolog Eiger (Egr) under *GMR-Gal4* control, which triggers apoptosis-dependent degeneration in the developing eye (Chen et al., 1999; Igaki et al., 2002; Ma et al., 2012; Walton and Gumucio, 2021). While *GMR>egr* animals exhibited a pronounced small-eye phenotype, *rpk* depletion led to a partial but statistically significant rescue of eye size (Fig. S2H). Conversely, enhancing depolarization, either by overexpressing the red light-activated depolarizing channelrhodopsin *ReaChR* or by depleting the Na⁺ pump α subunit (*Atpa*), accelerated the clearance of *scrib* mutant clones (Figs. 2G, 2H, S2F, S2G).

Collectively, these results identify membrane potential fluctuations as an electrophysiological regulator that tunes the competitive fitness of polarity-deficient cells during TSCC.

Membrane potential regulates *scrib*-mediated TSCC via Hh signaling activation

To further elucidate the mechanism by which membrane potential changes facilitate the clearance of *scrib* mutant clones, we performed gene set enrichment analysis (GSEA) of FlyBase signaling pathways on upregulated genes of *scrib*^{-/-} cells compared to WT cells. This revealed significant enrichment of the Hh signaling pathway (Fig. 3A), a key driver of tumorigenesis and tissue homeostasis regulation (Briscoe and Théron, 2013). Hh signaling is initiated when Hh ligand binding alleviates Patched (Ptc)-mediated repression of Smoothened (Emmons-Bell and Hariharan, 2021), stabilizing the activated form of the transcription factor Cubitus interruptus (Ci) and inducing target genes such as *decapentaplegic* (*dpp*) and *ptc* (Kim and Jain, 2020) (Fig. 3B). To examine Hh pathway activation during *scrib*-mediated cell competition, we analyzed Hh target expression. *scrib* mutant clones exhibited strong upregulation of Ptc protein levels (Figs. 3C and S3A) and *dpp-lacZ* transcription (Figs. 3D and S3B). Additionally, *scrib* clones also exhibited elevated membrane accumulation of Smo and increased expression of Ci protein levels (Figs. 3E and S3C), further confirming Hh signaling activation during *scrib*-mediated TSCC. Notably, widespread depletion of *scrib* in a non-mosaic context did not result in Hh pathway activation (Fig. S3D–S3F).

Next, we evaluated the functional role of Hh signaling activation in *scrib*-mediated TSCC by genetically perturbing the pathway. Activating Hh signaling via *ci* or *smo* overexpression or *ptc* KD further reduced *scrib* clone size (Fig. 3F and 3G) and enhanced *scrib* loss-induced apoptosis (Fig. S3G). Conversely, inhibition of Hh signaling via KD of *smo* or *ci* significantly impeded the elimination of *scrib* clones (Fig. 3F and 3G) and suppressed apoptosis (Fig. S3G). Of note, under physiological conditions, clones with increased Hh signaling activity exhibited

compromised competitive fitness (Fig. S3H), while Hh signaling inhibition has no significant effect on clone size (Fig. S3I). Together, these findings indicate that clonal activation of Hh signaling promotes a loser cell fate and is crucial for *scrib*-mediated TSCC.

Interestingly, recent evidence indicates that membrane potential changes can modulate Hh signaling in *Drosophila* epithelia (Spannl et al., 2020; Cong and Cagan, 2024). Therefore, we further performed genetic epistasis analysis to explore the hierarchy between membrane potential and Hh signaling in regulating *scrib*-mediated TSCC. Voltage normalization by knocking down *rpk* in *scrib* mutant cells dramatically blocked membrane accumulation of Smo (Figs. 3H and S4A), as well as upregulation of Ci (Fig. S4B), demonstrating that Hh pathway regulation is voltage-dependent during TSCC. Conversely, we tested whether Hh perturbations affect membrane depolarization. The increase in both membrane potential and intracellular calcium in *scrib*^{-/-} clones remained unaffected by the co-depletion of *ci* (Figs. 3I, S4C, S4D). These data indicate that Hh signaling functions downstream of membrane potential changes in regulating the elimination of *scrib* clones. To test whether membrane potential perturbation alone is sufficient to induce Hh activation in otherwise WT cells, we overexpressed *ReaChR* or knocked down *ATPa* or *rpk* in WT epithelial clones. We found that none of these manipulations significantly altered the expression levels of Ci, Ptc, or Smo (Fig. S4E). These findings indicate that while membrane depolarization is necessary for Hh activation in *scrib*-deficient clones, it is insufficient to initiate Hh signaling in the absence of additional oncogenic or competitive cues.

Collectively, our findings establish membrane potential as one of the upstream regulators of Hh signaling during *scrib*-mediated TSCC. Depolarization in *scrib*^{-/-} cells amplifies Hh pathway activity, which in turn drives their elimination, while voltage normalization rescues competitive fitness (Fig. 3J). This mechanistic link highlights the interplay between bioelectric cues and developmental signaling in cell competition.

A potential role of the V_m-Hh axis in *scrib*-mediated cell competition in mammalian MDCK cells

Cell competition is a widely observed mechanism that eliminates lower-fitness cells during embryonic development and maintains tissue homeostasis in both *Drosophila* and mammals (Qin et al., 2005; Norman et al., 2012; Wagstaff et al., 2016; Zheng et al., 2023; Wang et al., 2026). While studies in *Drosophila* have established *scrib* as a critical regulator of TSCC

(Norman et al., 2012; Kanda and Igaki, 2020; Kong et al., 2022; Melero et al., 2025), *Scrib* KD in Madin-Darby Canine Kidney (MDCK) cells has emerged as a key model for probing both common and divergent mechanisms of cell competition across species (Wagstaff et al., 2016; Tanimura and Fujita, 2020; Maruyama and Fujita, 2022). When co-cultured with WT MDCK cells, *Scrib*-KD cells are competitively eliminated (Maruyama and Fujita, 2022; Zheng et al., 2023) (Fig. 4A–4C), phenocopying outcomes observed in *Drosophila*. Leveraging this mammalian model, we investigated whether the V_m -Hh signaling axis plays a functionally analogous role in removing *Scrib*-deficient cells. We generated stable MDCK cell lines with KD of *Scrib*, *Scnn1g* (the mammalian homolog of *Drosophila rpk*), or both (Fig. S5A–S5C). After 4 days of co-culture at a 1:10 ratio (*Scrib*-KD: WT), the elimination of *Scrib*-KD cells was significantly suppressed by concurrent *Scnn1g* KD, as quantified by loser cell area and flow cytometric analysis (Figs. 4B–4E, S5D). Moreover, *Scrib*-KD loser cells also exhibited reduced mitochondrial activity (Figs. 4F, 4G, S5E), recapitulating findings in *Drosophila*. Collectively, these results suggest that the *Scrib*-depleted MDCK cells exhibit impaired mitochondrial activity and rely on V_m changes to regulate competitive elimination, pointing to a potential functional link for bioelectric signaling in cell competition.

Next, we evaluated whether V_m changes mediate Hh signaling activation in *Scrib*-KD MDCK cells. Mirroring our observations in *Drosophila*, co-culture conditions significantly upregulated transcription levels of Hh signaling target genes, including *GLI2*, *PTC1*, and *PTC2*, in *Scrib*-KD cells (Fig. 4H–4J), confirming Hh signaling activation in loser cells. Notably, the expression of *SMO* and *SHH* remained unchanged (Figs. 4K and S5F). However, double KD of *Scrib* and *Scnn1g* in MDCK cells co-cultured with WT cells significantly suppressed Hh signaling activation (Fig. 4H–4J), suggesting that reduced depolarization correlates with inhibited Hh signaling activation in *Scrib*-KD cells.

Collectively, these data reveal striking parallels between *Drosophila* and mammalian systems, where *Scrib*-KD-driven competitive interactions are associated with mitochondrial defects, which in turn appear to trigger V_m -dependent Hh signaling activation to facilitate clearance. This genetic interplay between bioelectrical homeostasis and developmental signaling pathways suggests a potentially conserved quality-control mechanism for cellular fitness and oncogenic surveillance, although further validation *in vivo* will be required to definitively establish evolutionary conservation.

Discussion

Epithelial homeostasis relies on robust mechanisms to eliminate aberrant cells and prevent tumorigenesis (Nagata and Igaki, 2018; Maruyama and Fujita, 2022; Melero et al., 2025). In this study, leveraging the scRNA-seq, *Drosophila* genetics, and mammalian cell cultures, we uncover a critical role for bioelectric signaling in tumor-suppressive cell competition (TSCC), demonstrating that membrane potential (V_m) depolarization drives the elimination of polarity-deficient *scrib* mutant cells via Hh pathway activation (Fig. 4L). Our findings bridge mitochondrial dysfunction, ion homeostasis, and developmental signaling into a unified framework that explains how cellular fitness is electrically encoded during oncogenic surveillance.

Mitochondrial ETC dysfunction drives bioelectric-dependent Hh signaling to orchestrate cell competition

Building on single-cell transcriptomic and functional analyses, we demonstrate that *scrib*^{-/-} cells exhibit mitochondrial ETC defects, predominantly affecting Complexes I and IV. This dysfunction diminishes ATP production, thereby impairing ATP-dependent ion transporters, including the Na⁺/K⁺-ATPase. The resultant plasma membrane depolarization creates a bioelectric “loser” state that sensitizes cells to competitive elimination. Importantly, genetic modulation of membrane potential, via *rpk* KD or *ReaChR* overexpression, was found to rescue or exacerbate the elimination of *scrib*^{-/-} clones, respectively, directly linking V_m dynamics to competitive fitness (Fig. 4L). These findings align with emerging evidence that mitochondrial metabolism and ion flux are intricately connected to cellular fitness under oncogenic stress (de la Cova et al., 2014).

A notable feature of our observations is that bioelectric and metabolic defects propagate throughout the entire *scrib* mutant clone, rather than being restricted to the boundary interface with WT neighbors. This uniform phenotype suggests that long-range mechanical forces and paracrine signaling may transmit boundary-induced stress into the clone interior, synchronizing the “loser” state across the tissue (Zheng et al., 2023). This observation reinforces the inherently non-cell-autonomous nature of cell competition: while *scrib*^{-/-} cells are eliminated when surrounded by WT neighbors, they hyperproliferate in homogeneous tissues where competitive pressure is absent (Chen et al., 2012). Our study corroborates this fundamental principle while revealing a specific mechanistic link between polarity loss, bioelectric alteration, and competitive elimination. Future investigations will be essential to dissect the precise molecular cues that initiate this cascade and to determine how these signals

are coordinated across the clone.

Furthermore, we identify Hh signaling as a key downstream effector linking V_m changes to TSCC. *scrib* mutant loser clones exhibited increased activation of the Hh pathway, which was functionally critical: inhibition of Hh signaling rescued *scrib*^{-/-} clone survival, whereas its activation accelerated elimination. Interestingly, Smo membrane accumulation exhibited partial voltage dependence, consistent with previous reports demonstrating that plasma membrane potential modulates Hh pathway activity (Spannl et al., 2020; Cong and Cagan, 2024). Collectively, these findings suggest that membrane depolarization and subsequent Hh signaling activation constitute a previously unrecognized, albeit partial, mechanism underlying the clearance of *scrib* mutant cells. Moreover, our results redefine the role of Hh signaling by extending its function beyond classical morphogen gradients and positioning it as a bioelectrical sensor of cellular fitness during oncogenic surveillance.

Non-apoptotic mechanisms and environmental cues regulating V_m in cell competition

The regulatory role of V_m in controlling cell proliferation extends well beyond excitable cells such as neurons. Dynamic changes in V_m are now recognized as critical drivers of cell cycle progression, volume control, and proliferation across virtually all cell types (Levin, 2014). Interestingly, our data show that KD of *rpk* significantly rescued *scrib*^{-/-} clone size without altering apoptosis levels (Fig. S2G). This finding suggests that membrane depolarization influences competitive fitness through mechanisms distinct from the suppression of apoptotic pathways. This observation aligns with emerging evidence indicating that V_m can modulate cell growth and proliferation across diverse biological contexts (Levin, 2021; Sasaki et al., 2025).

Multiple signaling networks intersect with V_m to regulate cellular fate. For instance, membrane depolarization has been shown to promote mitosis via the voltage-dependent activation of ERK signaling in human cells (Sasaki et al., 2025). Furthermore, recent work demonstrates that V_m mediates the cellular response to mechanical pressure by orchestrating Hippo and MAPK pathway signaling, thereby regulating epithelial cell density through the coordinated control of growth, proliferation, and elimination (Mukherjee et al., 2026). Given these complex interactions, future studies will be essential to delineate whether V_m also regulates the fate of “loser” cells by modulating growth rates or through other non-apoptotic mechanisms.

Beyond the downstream mechanisms characterized here, a pivotal unanswered question concerns the nature of the upstream signals from WT “winner” cells that initiate V_m depolarization in *scrib*^{-/-} “loser” cells. Our data clearly demonstrate that mitochondrial defects and membrane depolarization in *scrib*^{-/-} cells are contingent upon a competitive tissue environment; however, the specific mediators driving this response remain undefined. Potential candidates include mechanical compression exerted by surrounding proliferating cells, the secretion of specific paracrine factors, or direct competition for metabolic resources. Dissecting the relative contributions of these mechanisms will be essential for future studies to fully elucidate how the competitive microenvironment orchestrates bioelectric reprogramming in loser cells.

Functional parallels in mammalian epithelia and implications

The translational significance of our findings is suggested by the functional parallels observed for this bioelectric mechanism across mammalian systems. In *Scrib*-KD MDCK cells, we observed mitochondrial dysfunction and activation of the Hh pathway, phenotypes that closely mirror those in *Drosophila* epithelia lacking *scrib*. Crucially, *Scnn1g* (ENaC- γ) KD normalized V_m and suppressed Hh activation in *Scrib*-KD cells, confirming the functional interplay between ion flux and developmental signaling. These findings point to a potential role of bioelectricity in cellular fitness assessment and suggests that V_m modulation could be therapeutically targeted to enhance oncogenic surveillance across species.

While the Hh signaling pathway is canonically associated with tumor promotion in numerous cancers (Rubin and de Sauvage, 2006), our study reveals a paradoxical tumor-suppressive function of Hh activation in clearing *scrib* mutant precancerous cells. This context-dependent duality in Hh signaling is not unprecedented. In *Drosophila* eye imaginal discs, excessive Hh pathway activation has been shown to induce apoptosis: knocking down *ci* suppresses Hid-induced apoptosis and alleviates the small-eye phenotype, whereas overexpression of *ci* or its upstream activator *smo* enhances apoptosis (Liu et al., 2021). Our recent work further supports this nuanced role of Hh signaling in tumorigenesis (Kong et al., 2025): when constitutively active Ci (*ci*^{ACT}) is co-expressed with activated Yorkie (*yki*^{S168A}), a key transcriptional co-activator of the Hippo pathway, the resulting clones gain a competitive growth advantage and progress to malignant tumors (Kong et al., 2025). This synergistic oncogenesis underscores that the functional outcome of Hh activation is critically dependent on the broader signaling landscape and cooperating genetic lesions.

Clinically, these findings suggest that pharmacological agents targeting membrane potential (e.g., ion channel modulators) or Hh activity could be repurposed to enhance cell competition in precancerous tissues. However, critical questions remain: How do depolarization-sensitive effectors interface with canonical Hh pathway components? Do other oncogenic insults, such as oncogenic *Ras* activation, converge on similar bioelectric checkpoints? Addressing these questions will require advanced optogenetic tools to spatiotemporally manipulate V_m *in vivo* and high-resolution imaging methods to elucidate Smo dynamics in depolarized membranes of loser cells.

Notably, during the revision of this manuscript, a recent study demonstrated that in crowded epithelial tissues, cells with the lowest energy levels and membrane potential are selectively extruded. This process is mediated by mechanical forces and bioelectric signals, particularly through ENaC and Piezo1 channels, which coordinate to eliminate less-fit cells, thereby maintaining healthy epithelial turnover and preventing overcrowding or potential neoplastic transformation (Mitchell et al., 2025). Given that both cell competition and cell extrusion, collectively termed epithelial defense against cancer (EDAC), contribute to tumor suppression, our findings, combined with this emerging work, suggest a potentially conserved and integrative role for bioelectrical signaling in cellular fitness surveillance. We propose that changes in V_m act as a regulatory layer, potentially upstream or complementary to established EDAC pathways, particularly in contexts involving altered ion channel expression or membrane excitability.

The mammalian evidence presented in our study is derived exclusively from co-culture experiments using MDCK cells. The MDCK *Scribble* KD model, originally established by Norman et al., has since been adopted by numerous independent studies and serves as a widely accepted *in vitro* standard for investigating cell competition in mammals (Norman et al., 2012). However, we acknowledge that this simplified system cannot fully recapitulate the architectural complexity of intact mammalian epithelia or the dynamic *in vivo* microenvironment. While our data reveal intriguing functional parallels between *Drosophila* and MDCK cells regarding the V_m -Hh signaling axis, we caution that the evolutionary conservation of this mechanism in mammals remains to be fully validated. Future studies employing more physiologically relevant systems, such as mouse models and organoids, will be essential to confirm whether these parallel functions represent a deeply conserved evolutionary strategy.

In summary, by integrating *Drosophila* genetics, mammalian epithelial models, and single-cell transcriptomics, we have identified a potentially conserved mechanism whereby mitochondrial dysfunction drives membrane depolarization, which in turn licenses Hh-mediated elimination of polarity-deficient cells. This work advances our understanding of bioelectricity as a fundamental regulator of tissue homeostasis and provides a roadmap for leveraging electrophysiological pathways in cancer prevention strategies. The unexpected role of membrane potential in tumor suppressive cell competition uncovered here highlights the need for further mechanistic studies across diverse cell competition models, which might inform innovative therapeutic approaches for cancer management.

Materials and methods

Drosophila strains

All *Drosophila* stocks and crosses were raised on standard food at 25°C, unless otherwise specified. The composition of the standard cornmeal yeast food was as follows: 50 g of corn flour, 9 g of agar, 24.5 g of dry yeast, 7.25 g of white sugar, 30 g of brown sugar, 4.4 mL of propionic acid, 12.5 mL of ethanol, and 1.25 g of nipagin per liter. *Drosophila* strains used are listed in Table S2.

Single-cell RNA sequencing

The scRNA-seq for *Drosophila* eye disc tumors has been previously described (Zhao et al., 2025). Briefly, 150 eye-antennal discs from late third instar larvae were obtained for both WT and *scrib*-deficient groups (three replicates each). The discs were dissected in 1× PBS solution and digested using sCellLive™ Tissue Dissociation Mix (Singleron). Cell viability was assessed using Acridine Orange/Propidium Iodide (AO/PI) fluorescence staining (#RE010212, Countstar). Samples with cell viability exceeding 90% were adjusted to a concentration of $4.0\sim 5.0 \times 10^5$ cells/mL for library construction using GEXSCOPE® Single Cell RNA Library Kits (Singleron) (Dura et al., 2019). Qualified single-cell suspensions were loaded onto the microwell chip using the Singleron Matrix® Single Cell Processing System (Singleron) along with barcoding beads. The cells were lysed to capture mRNA using the barcoding beads, followed by reverse transcription to synthesize cDNA. The amplified cDNA was fragmented and ligated with sequencing adapters, and libraries that passed quality control were sequenced on the Illumina NovaSeq 6000 with 150 bp paired-end reads.

Analysis of scRNA-seq data

The generated single-cell FASTQ files were analyzed using Celescope (v1.10.0) software with standard pipelines (<https://github.com/singleron-RD/CeleScope>) against *Drosophila melanogaster* reference genome (*Drosophila_melanogaster*.BDGP6.32.108) with exogenous gene sequence (GFP and Gal80) inserted. The filtered gene expression matrices from the Celescope pipelines were further analyzed in Seurat v5.1.0 (Hao et al., 2024). We first retained high-quality cells that expressed a total number of genes between 200 and 5000 and expressed low percentage of mitochondrial genes (<30% for *scrib*-deficient group and <40% for WT group). For each batch, doublet was predicted by DoubletFinder (v2.0.3) using the top30 PCs (McGinnis et al., 2019). Optimal PK value was determined by the find.pK function for each scRNA-seq dataset. Assuming 15% doublet rate, 13,949 high-confidence doublets were found and filtered for downstream analysis. Then scRNA-seq datasets were integrated across batches using Seurat integration procedure. Integrated dataset was then scaled with regression using mitochondrial genes and total count RNA. The data was then reduced using UMAP with the top 60 dimensions. The cell clusters were determined using the FindNeighbors and FindClusters functions in Seurat. Using known markers, hemocytes and glia cells were removed from the downstream analysis. To identify *scrib*^{-/-} cells, scRNA-seq datasets from *scrib*-deficient groups were integrated. The clusters of *scrib*^{-/-} cells were annotated based on the expression level of GFP. Differentially expressed genes between the clusters of *scrib*^{-/-} cells and WT cells were identified with the function *FindMarkers()*, using the default log-fold change threshold value and a minimum percentage of cells in which the gene is detected of 25%. To identify enrichment of particular gene groups or pathways, differentially expressed gene lists were subject to PANGEA using the KEGG and FlyBase signaling pathway sets (Hu et al., 2023).

Live imaging

Larvae were washed with PBS prior to dissection. Live tissue was dissected in Schneider's media. DiBAC4(3) staining was performed as previously described (Chen et al., 2012). The eye discs were incubated in 1.9 µM DiBAC4(3) (bis-[1,3-dibutylbarbituric acid] trimethine oxonol) with Schneider's media for 10 min and discs were imaged right away. For TMRM staining, the eye discs or cells were incubated in 1 µM TMRM with Schneider's media for 45 min at 37°C, washed with PBS. The imaged using Olympus FV3000. For calcium ion staining experiments, larvae were dissected and cleaned in Schneider's medium, and incubated with 3

μM Fluo-4-AM in Schneider's media for 45 min at 37°C, washed with PBS two times. The samples were imaged using Zeiss Axio Observer with ApoTome2. Chemicals are listed in the Table S2.

Modeling cell competition in *Drosophila* eye imaginal discs

For tumor suppressive cell competition, fluorescently labeled mitotic clones were produced in the eye-antennal discs using the mosaic analysis with a repressible marker (MARCM) system with the following tool strains: *y, w, ey-Flp1; act>y+>GAL4, UAS-GFP; FRT82B, Tub-GAL80* (*ey-FLP1* 82B tester). Larvae were then allowed to grow at 25°C until third-instar stage and clone areas were measured using ImageJ, for clones in the eye-antennal disc. For *Myc*-induced cell competition, the fluorescently labeled mitotic clones were produced in the eye-antennal discs with the following tool strains: *y, w, hsFlp1.22; tub>CD2>Gal4/UAS-GFP* (for control) and *y, w, hsFlp1.22; tub>myc>Gal4, UAS-GFP* (for competition). First instar larvae were transferred to food supplemented with fresh yeast paste and allowed to develop at 25°C. To induce clones, second instar larvae were heat shocked at 37°C for 30 min. Then larvae were allowed to grow at 25°C until 48 hours.

Immunofluorescence staining and imaging

The third-instar larvae were dissected in PBS and fixed for 30 min in 4% PFA at room temperature, washed with PBS containing 0.1% Triton X-100 solution (PBST) three times. Then the samples were blocked in 5% normal goat serum for 1 h at 4°C. The samples were incubated with primary antibodies (1:200) in antibody diluent overnight at 4°C; washed with PBST three times, samples were then incubated with secondary antibodies (1:800) at room temperature for 2 h. After three PBST washes, samples were mounted with DAPI-containing medium (Thermo Fisher Scientific, Cat# S36939) and imaged using Olympus FV3000. Images were processed with ImageJ. Antibodies are listed in the Table S2.

Mammalian cell culture

Madin-Darby canine kidney (MDCK) cells were cultured as previously described (Qin et al., 2005; Norman et al., 2012). These cells were a kind gift from Thuan Beng Saw (Westlake University). Cells were cultured in DMEM (containing GlutaMAX, high glucose and pyruvate; Life Technologies) supplemented with 10% fetal bovine serum (FBS; Life Technologies) and 1% penicillin-streptomycin (Life Technologies) at 37°C with 5% CO₂. To

construct shRNA vectors to KD the canine *Scrib*, partial sequences were obtained by RT-PCR from MDCK cells and screened for candidate small interfering RNA sequences using rational design criteria. Target sequences *Scrib*-KD1 and *Scrib*-KD2 sufficiently suppressed *Scrib* expression. Sequences of the *Scrib*-KD sense oligonucleotides are listed in the Table S2. Sense and antisense oligonucleotides for shRNA were cloned into the HpaI and XhoI sites of pLL3.7, and pLL3.7.GFP was used as a negative control. To construct pLV.mcherry *Scnn1g* shRNA, sense and antisense oligonucleotides for shRNA were cloned into BamHI and EcoRI sites of pLV.mcherry (Oligoengine). Selection was carried out with MA900 Multi-Application Cell Sorter (Sony Biotechnology). The KD efficiency in MDCK cells was confirmed by Western blotting.

Mammalian cell competition assays and sorting

Cell competition assays on MDCK cells were conducted in 6-well plates. Control cultures were established at the center of the well at a ratio of 1 *Scrib* KD:10 WT cells. Live imaging commenced 96 hours after the final media change, with regular media exchanges every 2 days. Imaging was performed using Olympus FV3000 and Leica Thunder microscopes, and the acquired images were analyzed with ImageJ and Photoshop CS5 (Adobe) for merging and resizing. WT cells, with or without GFP, and WT cells co-cultured with *Scnn1g*-KD, *Scrib*-KD, or *Scrib*-KD + *Scnn1g*-KD, were plated in 6-well dishes at a ratio of 10 (winner):1 (loser). After 4 days, the co-cultured cells were harvested and washed twice with ice-cold PBS. Subsequently, the cells were digested with Trypsin for 8 minutes at 37°C, followed by another PBS wash to prepare a single-cell suspension. The suspension was sorted into four groups using the MA900 Multi-Application Cell Sorter (Sony Biotechnology): winner cells (GFP⁻), loser cells (GFP⁺), loser cells (mCherry⁺), and loser cells (mCherry⁺ GFP⁺). The data were analyzed using FlowJo software (version 10).

Western blotting

The harvested cells were washed with 1 mL cold PBS and lysed in 200 µl NP40 buffer. After the samples were rotated for 40 min at 4°C, lysates were centrifuged at 13000 rpm for 13 min. Then the lysates were mixed with 5× sample loading buffer and heated at 95°C for 7 min. After separation of proteins by SDS-PAGE, the proteins were transferred to PVDF membrane (Merck Millipore NO. IPVH00010). The membrane was blocked with 5% milk-PBST for 1 h, incubated with primary antibody overnight at 4°C, washed with PBST, incubated with

secondary antibody at room temperature for 1 h, and washed with PBST four times. Proteins were detected using the Vilber Fusion FX6, and antibodies are listed in the Table S2.

RNA extraction and RT-qPCR

MDCK cells total RNA was prepared using an RNA preparation kit. The yield and quality of the RNA were validated by electrophoresis and by using a NanoDrop spectrophotometer (Thermo Fisher Scientific) to measure the concentration and 260/280 nm absorbance ratio. cDNA was produced using FastKing gDNA dispelling RT SuperMix. Quantitative PCR was performed with SYBR Green Realtime PCR Master Mix and the relative abundance of mRNA was calculated using comparative Ct method.

Quantification and statistical analysis

To quantify the small-eye phenotype induced by *GMR>egr*, the eye area of female flies was measured manually using ImageJ. To quantify the size of clones derived from discs, images obtained through fluorescence microscopy were analyzed using ImageJ. For clonal analysis in imaginal discs, fluorescence microscopy images were used to determine clone size, calculated as the ratio of the total GFP-positive clone area to the total disc area (expressed as a percentage) for each genotype. Statistical analyses were conducted using GraphPad Prism 8.0. Data are presented as mean $\pm$ standard deviation (Swanton et al., 2024). Comparisons between two groups were performed using an unpaired two-tailed Student's *t*-test, assuming normal distribution and homogeneity of variance. Comparisons among three or more groups were conducted using one-way analysis of variance (ANOVA) followed by appropriate post-hoc tests. The number of animals used per experiment is specified in the figure legends. A *P*-value < 0.05 was considered statistically significant. Statistical significance is denoted as follows: ns, not significant; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$.

Data availability

All data generated in this study are available within the article and/or SI Appendix. The scRNA-seq datasets for the WT and *scrib*-deficient groups have been deposited in the NCBI BioProject database under the accession numbers PRJNA1064441 and PRJNA1261227, respectively.

CRediT authorship contribution statement

Kun Zhao: Writing - Original draft, Review & Editing, Formal analysis, Data curation, Investigation, Visualization. **Dian Lv:** Writing - Original draft, Review & Editing, Formal analysis, Data curation, Investigation, Visualization, Software. **Du Kong:** Formal analysis, Data curation, Investigation, Visualization. **Yifan Guo:** Formal analysis, Data curation, Investigation, Visualization, Software. **Xianping Wang:** Formal analysis, Data curation, Investigation, Visualization. **Sihua Zhao:** Investigation. **Jiandong Zheng:** Investigation. **Jia Jing:** Investigation. **Sha Song:** Investigation. **Jingjie Mu:** Investigation. **Yixin Wu:** Investigation. **Xihui Chen:** Investigation. **Xuan Guo:** Resources, Methodology. **Yunpeng Huang:** Resources, Methodology. **Wenyan Xu:** Investigation. **Min Jiang:** Resources, Methodology. **Xiaoqin Li:** Writing - Original draft, Review & Editing, Formal analysis, Data curation, Investigation, Visualization, Conceptualization, Supervision, Project administration, Validation. **Xianjue Ma:** Writing - Original draft, Review & Editing, Conceptualization, Supervision, Project administration, Validation.

Conflict of interest

The authors declare no competing interests.

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Figures legends

Fig. 1. *scrib* mutant clones exhibit decreased mitochondrial activity. **A:** Experimental design of
 scRNA-seq performed in WT and *scrib* mutant clone bearing eye-antennal discs. **B:** UMAP
 representation of WT and *scrib*^{-/-}/WT eye-antennal disc cells. Each cluster represents a cell
 type. **C:** Dot plot showing the expression of top markers for each cell type in the larval
 eye-antennal discs. **D:** UMAP cluster plot of cells from *scrib*^{-/-}/WT larval eye-antennal disc
 (after removing hemocytes, glia cells, and unknown clusters). Feature plot shows the
 expression of *GFP*, *Mmp1*, and *scaf*. **E:** KEGG Gene enrichment analysis of upregulated
 genes of *scrib*^{-/-} clusters compared to cells from WT eye-antennal disc. **F:** Heatmap showing
 scores of mitochondria complexes throughout *scrib*^{-/-} clusters and cells from WT eye
 antennal disc. **G:** Fluorescent images of live eye discs expressing GFP-labeled clones of WT
 and *scrib*^{-/-} incubated with 100 nM TMRM. **H:** Fluorescent micrographs of eye discs bearing
 GFP-labeled WT or *scrib*^{-/-} clones to examine the *in vivo* production of ATP using
 ATP-SPARK. Quantification of ATP generation of indicated genotypes as shown (mean ± SD;
 unpaired *t* test). **, *P* < 0.01. **I:** Schematic diagram of disrupted mitochondrial ETC core

components in *scrib* mutant clones. Scale bars, 10 μ m (**G**); 20 μ m (**H**). scRNA-seq, single-cell RNA sequencing; WT, wild-type; EAB, eye-antennal border; MF, morphogenetic furrow; PM, peripodial membrane; Pro&Pre, photoreceptor progenitors and photoreceptor precursors; EPR, early photoreceptors; LPR, late photoreceptors; INT, interommatidial cells; SMW, second mitotic wave; PG, perineurial glia; WG&SPG, wrapping glia and subperineurial glia; TMRM, tetramethylrhodamine methyl ester; SPARK, separation of phases-based activity reporter of kinase; SD, standard deviation.

Fig. 2. Plasma membrane depolarization facilitates *scrib* cell competition. **A:** Schematic diagram illustrating ATP-mediated modulation of Na^+/K^+ -ATPase activity to alter membrane potential. **B:** Fluorescent micrographs of eye disc bearing GFP-labeled WT or *scrib*^{-/-} clones stained with anti-ATP α antibody. Relative ATP α fluorescence intensity is quantified on the right (mean $\pm$ SD; unpaired *t* test). ***, *P* < 0.001. **C:** Volcano plot showing differential expressed genes between *scrib*^{-/-} clusters and WT eye-antennal disc. **D:** Fluorescent micrographs of eye disc bearing GFP-labeled WT or *scrib*^{-/-} clones were stained with anti-ENaC γ antibody. Relative ENaC γ fluorescence intensity is quantified on the right (mean $\pm$ SD; unpaired *t* test). ***, *P* < 0.001. **E:** Fluorescent images of live eye discs expressing GFP-labeled clones of WT and *scrib*^{-/-} incubated with DiBAC4(3). Relative DiBAC4(3) fluorescence intensity is quantified on the right (mean $\pm$ SD; ordinary one-way ANOVA). ***, *P* < 0.001. **F:** Eye-antennal discs bearing *ey*-Flp-MARCM-induced GFP-labeled clones with indicated genotypes and quantification of the relative size of GFP-labeled clones with indicated genotypes (right) (mean $\pm$ SD; Brown-Forsythe and Welch ANOVA tests). ns, not significant; *, *P* < 0.05; **, *P* < 0.01; ****, *P* < 0.0001. Scale bars, 20 μ m (**B**, **D**, **E**); 100 μ m (**F**). WT, wild-type; SD, standard deviation; MARCM, mosaic analysis with a repressible marker

Fig. 3. Membrane potential regulates *scrib*-mediated TSCC via Hh signaling activation. **A:** scRNA-seq analysis revealed significant enrichment of the Hh signaling pathway in *scrib*^{-/-} cells compared to WT cells. **B:** Simplified diagram of Hh signaling pathway activation in

Drosophila. **C–E**: Eye-antennal discs bearing *ey*-Flp-MARCM-induced GFP-labeled WT and *scrib*^{-/-} clones stained with anti-Ptc antibody (**C**), anti- β -galactosidase antibody (**D**), and anti-Smo antibody (**E**). **F**: Eye-antennal discs bearing *ey*-Flp-MARCM-induced GFP-labeled clones with indicated genotypes. **G**: Quantification of the relative size of GFP-labeled clones with indicated genotypes (mean $\pm$ SD; Brown-Forsythe and Welch ANOVA tests). **, $P < 0.01$; ****, $P < 0.0001$. **H**: Eye-antennal discs bearing *ey*-Flp-MARCM-induced GFP-labeled WT and *scrib*^{-/-} clones stained with anti-Smo antibody. **I**: Fluorescent images of live eye discs expressing GFP-labeled clones of WT, *scrib*^{-/-}, *ci*^{RNAi}, and *scrib*^{-/-} + *ci*^{RNAi} incubated with DiBAC4(3). **J**: Cartoon illustration of how depolarization in *scrib*^{-/-} cells activates Hh signaling, which in turn drives the clearance of *scrib*^{-/-} cells. Scale bars, 50 μ m (**C**, **D**, **E**, **H**, **I**); 100 μ m (**F**). Hh, hedgehog; WT, wild-type; MARCM, mosaic analysis with a repressible marker; SD, standard deviation.

Fig. 4. A parallel role of V_m -Hh axis in regulating *Scrib*-KD-induced cell competition. **A**: Schematic of the assay for *Scrib*-KD-induced cell competition, encompassing four co-culture conditions. **B**: Co-culture images of GFP and WT MDCK, *Scnn1g*-KD and WT MDCK, *Scrib*-KD and WT MDCK, and *Scrib*-KD + *Scnn1g*-KD and WT MDCK cells, cultured at a ratio of 1:10 for 4 days. **C**: Quantification of the area occupied by loser cells (GFP-positive) from the images in (**B**) (mean $\pm$ SD; Ordinary one-way ANOVA). ns, not significant; **, $P < 0.01$; ****, $P < 0.0001$. **D**: Flow cytometry analysis to determine the percentage of winner/loser cells after 4 days of co-culturing, following the same procedures as in (**A**) and (**B**). **E**: Quantifications of the percentage of loser cells (GFP-positive) from the data in (**D**) (mean $\pm$ SD; Ordinary one-way ANOVA). **, $P < 0.01$; ****, $P < 0.0001$. **F**: Confocal microscopy images of MDCK-GFP or MDCK-*Scrib*-KD cells co-cultured with WT MDCK cells and stained with 100 nM TMRM. **G**: Quantification of TMRM fluorescence intensity in loser cells (GFP-positive) from (**F**) (mean $\pm$ SD; Ordinary one-way ANOVA). ns, not significant; **, $P < 0.01$. **H–K**: qPCR analysis of *Gli2* (**H**), *Ptc1* (**I**), *Ptc2* (**J**), and *Smo* (**K**) expression levels in flow cytometry-sorted MDCK cells under pure culture or co-culture conditions (mean $\pm$ SD; Ordinary one-way ANOVA). ns, not significant; ***, $P < 0.001$; ****, $P < 0.0001$.

896 $P < 0.0001$. **L**: Working model depicting how membrane depolarization activates Hh signaling
897 to promote *scrib*-loss-induced tumor-suppressive cell competition. Scale bars: 50 μm (**B** and **F**
898 bottom panels); 100 μm (**F** upper panels). WT, wild-type; KD, knockdown; SD, standard
899 deviation; TMRM, tetramethylrhodamine methyl ester.







